



Face anti-spoofing with generated near-infrared images

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Abstract

Reflection differences between live faces and spoof faces under near-infrared spectrum make near-infrared image based methods obtain superior performance for face anti-spoofing. However, for conventional face recognition systems, near-infrared image based methods need additional near-infrared equipment to capture the input near-infrared images. In this paper, we propose a novel face anti-spoofing method which exploits the clues in both visible light (VIS) images and near-infrared (NIR) images without utilizing any near-infrared equipment during testing. Specifically, we first propose a novel multiple categories image translation generative adversarial network (MCT-GAN) which generates corresponding NIR images for VIS live and spoof face images. Then we utilize convolution neural network to learn fusing features from both VIS images and corresponding generated NIR images for the goal of live and spoof face classification. Qualitative and quantitative experiments demonstrate that our method obtains excellent results compared to the state-of-the-art methods.

Keywords Face anti-spoofing · Feature fusing · Generative adversarial network · Image translation

1 Introduction

Face recognition systems have been widely used in individual identification applications. Conventional face recognition systems usually do not take into account the attacks from spoof faces such as printed photos, digit photos, face videos and masks tailored to live faces. This results in the systems are vulnerable to the attacks [2]. Face anti-spoofing aims to distinguish spoof face attacks from live face accesses and make the spoof faces fail verification. It becomes a crucial step to improve the safety and reliability of a face recognition system.

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Face anti-spoofing has been a challenging task for a few years. Many related investigations have been published. Common face recognition systems leverage visible light (VIS) cameras to capture input images. As a consequence, most of previous face anti-spoofing methods focus on exploring discriminative clues from VIS images for live and spoof face classification. The early methods use handcrafted features to extract the differences between live and spoof faces in terms of texture [8, 10, 14, 35], image quality [17, 51], and vital sign [27, 46, 49]. For example, one of the most famous methods [8] extracts five low-level features (LBP, LPQ, CoALBP, BSIf and SID) from images in HSV and YCbCr color spaces for live and spoof face classification. It jointly exploits the color and texture information from the luminance and chrominance channels to improve the performance of face anti-spoofing. Wen et al. [51] extract the specular reflection, blurriness, chromatic moment, and color diversity features to describe the image distortions existing in spoof face images. It analyzes the differences caused by reflection properties of different materials and obtains better generalization performance. With the development of deep learning, deep neural networks are exploited to learn deep features from different-scale VIS images in RGB, HSV and YCbCr color spaces [5, 53]. The learned high level semantic features significantly boost the performance of face anti-spoofing. VIS images contain rich classification clues for face anti-spoofing, and significant improvements have been achieved by exploring classification clues from VIS images. However, most of previous methods still suffer from unsatisfied inter-scenario generalization performance because of the differences in materials of spoof faces, image quality, image acquisition equipment, and illumination conditions [9, 13, 30]. The trained models usually fit well on the training set, but the performance will dramatically drop if the testing set comes from a different scenario than the training set.

In recent years, some researchers turn to near-infrared images (NIR images) to find discriminative and generalized classification features. Yi et al. [54] exploit the texture differences in NIR images via extracting Gabor features from cropped NIR face images. It obtains excellent performance on their collected spoofing dataset. Sun et al. [45] first collect near-infrared differential images via turning on and off the active near-infrared lights, then analyze the pixel consistency between face and non-face areas and pixel intensities distribution of the face area in the near-infrared differential images to distinguish spoof faces from live faces. It fuses the context and texture information based on the reflection differences preserved in the near-infrared differential images to improve the performance. Compared with visible spectrum, the reflection differences between live faces and spoof faces are more obvious under near-infrared spectrum and they are able to provide more effective and generalized features for face anti-spoofing [25, 45, 54, 57]. However, most of face recognition systems use only visible light cameras to capture input images. The near-infrared image based face anti-spoofing approaches have to add additional near-infrared cameras. This not only means expensive hardware costs but also is extremely inconvenient for reality business scenes. After all, remodeling hardware of an existing system is a tricky and time-consuming project.

In order to take advantage of NIR images without adding any NIR devices, we propose to translate the VIS images to NIR images for face anti-spoofing by an image translation model, as shown in Fig. 1. It is more convenient and cheaper to apply such method to real businesses. The popular Generative Adversarial Networks (GANs) (e.g., CycleGAN [60], pix2pix [22]) have shown impressive results in image translation. These methods are well done in learning to translate an image from a source style to a target style using a training set of unpaired images or paired images. However, these methods do not distinguish the categories of the source and target style images, and just translate different categories of

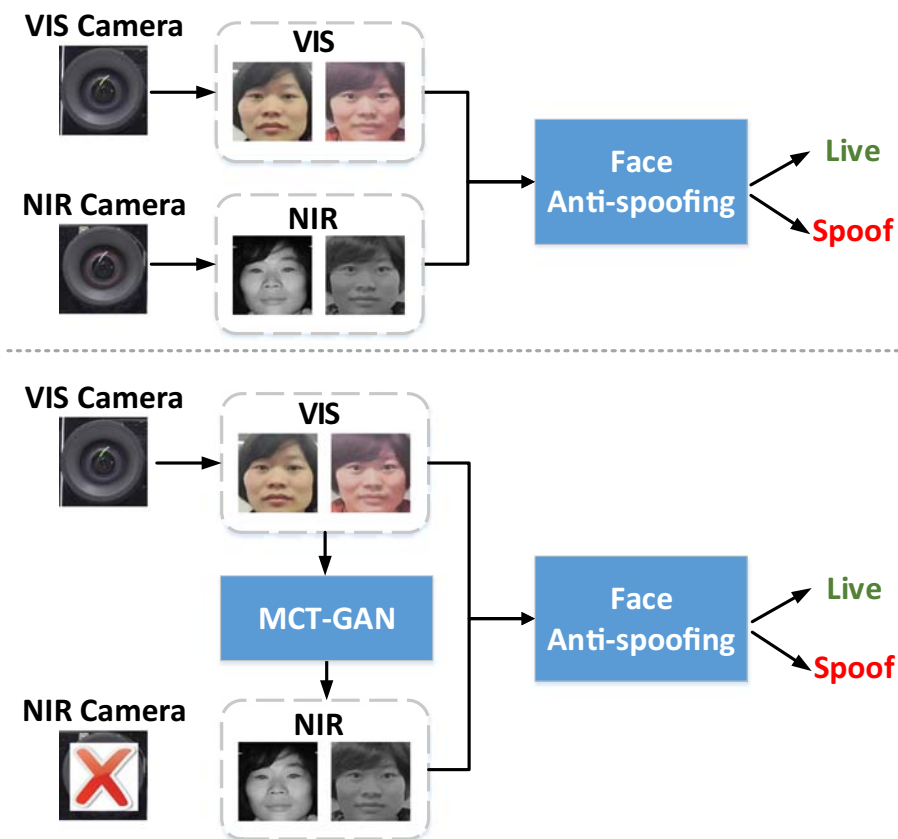


Fig. 1 Conventional near-infrared image based face anti-spoofing approaches utilize near-infrared camera to get near-infrared images, which means that additional near-infrared camera is needed. However, this work gets near-infrared images for face anti-spoofing only based on an existing visible light camera. It generates near-infrared live and spoof face images via a MCT-GAN

source style images to target style images with similar characteristics. Taking the task of translating horses to zebras as an example, they map a horse to the look of a zebra, but do not consider to map different breeds of horses to different breeds of zebras.

The NIR images of live and spoof faces have different characteristics, as shown in Fig. 2. For example, the skin of live faces looks more smooth than spoof faces. The eyes of live faces look a little turbid, but the eyes of spoof faces look clear. The background of live faces is almost black, but the background of spoof faces is clearly visible. There are some specular reflections existing in NIR images of spoof faces because of the materials of the spoof face surface. These are important clues we expect to utilize for face anti-spoofing. In order to improve the performance of face anti-spoofing via generated NIR images, the differences between NIR live and spoof face images should be preserved while the image translation model learns to translate the VIS face images to NIR face images. In other words, the image translation model should own the ability of giving source style (VIS) images different target style (NIR) according to their categories (e.g. live face images and spoof face images). To do that, we first propose a multiple categories image translation generative adversarial network,

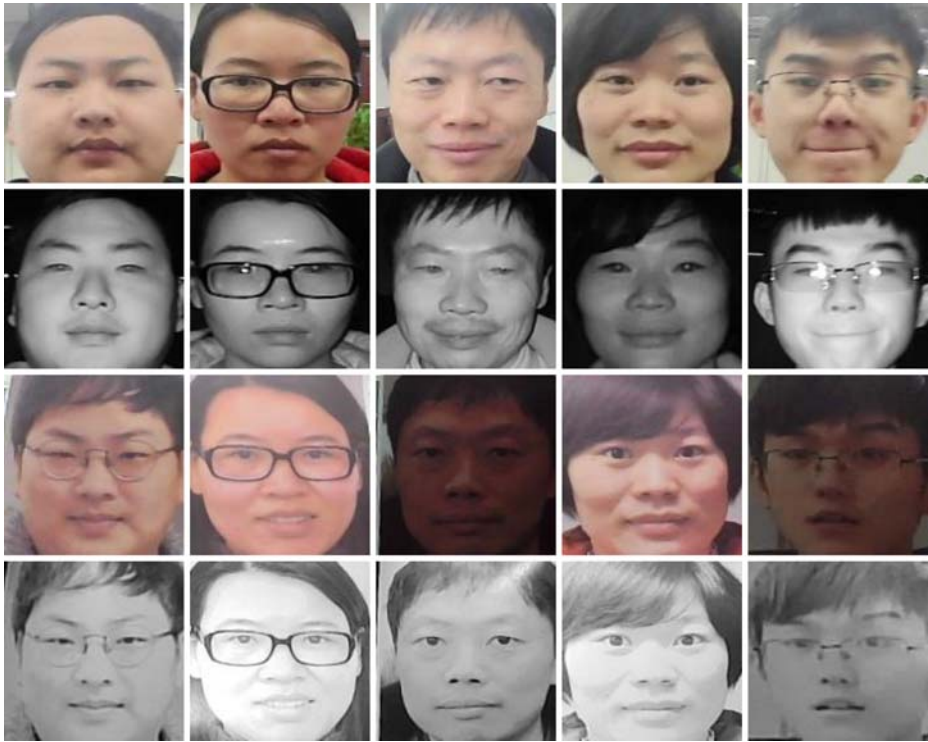


Fig. 2 Examples of VIS and NIR image pairs. Each column from top to bottom are VIS face image of live access, NIR face image of live access, VIS face image of print attack, NIR face image of print attack, respectively

which generates the NIR-style face images of VIS face images, meanwhile preserves the characteristics of NIR live face and NIR spoof face in a single model. Then, in order to take full advantage of information from original VIS and generated NIR face images for face anti-spoofing, we concatenate a pair of VIS and generated NIR face images as a learning sample to a CNN model. At last, the CNN model will learn a fusing representation of the paired images, which would further improve the performance of face anti-spoofing.

Overall, our contributions are as follows:

1. We propose a novel face anti-spoofing method which combines the discriminative ability of both visible light images and near-infrared images without utilizing any near-infrared equipment during testing.
2. We propose a novel multi-category image translation GAN (MCT-GAN) which generates the NIR-style face images of VIS face images, and preserves the near-infrared characteristics of live and spoof faces as well.
3. We do extensive qualitative and quantitative experiments and achieve excellent results compared with the state-of-the-art methods on two public datasets CASIA-MFSD [59] and Replay-Attack [11].

The rest of the paper is organized as follows: Section 2 reviews the related work. Section 3 details the algorithms of multiple categories NIR image generation and fusing feature learning. Section 4 summarizes the training and application procedures of the proposed

method. Section 5 presents our qualitative and quantitative experiment results and related analysis. Section 6 draws conclusions about our work.

2 Related work

Our work is to do face anti-spoofing based on generated NIR images. It is closely related with face anti-spoofing and image translation. In this section, we will first review prior face anti-spoofing methods in five groups: texture based methods, image quality based methods, vital signals based methods, deep learning based methods, and other hardware based methods. Then we review the related work in image translation.

2.1 Face Anti-spoofing

Texture based methods In the previous studies, face anti-spoofing is usually regarded as a binary classification problem where an input sample will be classified as one of the two classes: live or spoof. More than 66% of the face anti-spoofing methods try to find texture differences in visible light images to do live and spoof face classification [43]. The material, structure and manufacturing process of spoof faces are different from that of live faces, so the textures of most spoof faces and live faces are different. Early texture based methods focus on analyzing the luminance information of face images for gray-scale texture feature extraction. Multi-scale local binary patterns (LBP) [38] are utilized to capture texture differences in printing quality defects, specular reflections and shades to detect the face in front of the camera is a live face or a printed photo in [35]. It is a method which is easy to implement and computationally fast. Chingovska et al. [11] investigate the efficiency of four variants of LBP for distinguishing attacks of printed photos, displayed photos and displayed videos from live accesses. The regular LBP shows the best performance and has the lowest complexity. Excepting LBP and the variants of LBP, some of the other low level descriptors like local phase quantization (LPQ), histogram of oriented gradient (HOG), difference of gaussians (DOG) are also used to extract texture and edge information for improving the performance of live and spoof face classification [52, 59]. Gray-scale texture based methods have high demand for image quality so that they can get clear enough texture. Moreover, they ignore the differences in color which are also important clues for face anti-spoofing [7]. Boulkenafet et al. [8] extract color texture features from images in YCbCr and HSV color spaces via five descriptors: LBP, LPQ, BSIF, co-occurrence of adjacent local binary patterns (CoALBP), and scale-invariant descriptor (SID). The experiment results show that the extracted color texture information from luminance and chrominance channels is helpful to improve the performance of face anti-spoofing. Unfortunately, the differences in illumination conditions, acquisition equipment, spoofing medium, manufacturing process lead to differences in textures even for the same sample. Texture based methods are usually easily overfitted to the particular data conditions of training data and need better generalization ability [9, 13, 30].

Image quality based methods The image quality of many spoof faces has a certain distance from that of live faces after secondary imaging. Twenty-five general image quality measures are selected to extract the differences in image quality for face anti-spoofing [17]. Low complexity helps the method to run fast and be user-friendly. Generally speaking, there are specular reflections in many spoof face images because of the surface materials of spoof faces. Some images of print attacks and display attacks look blurry. The color diversity of

spoof faces heavily depends on the producers of spoof faces like printers and electronic equipment, which leads to color distortion of spoof face images. Wen et al. [51] analyze the image distortions in terms of specular reflection, blurriness, chromatic moment, and color diversity to distinguish spoof face images from live face images. It analyses the global reflection differences of the face images and obtains better generalization performance. The detected faces are segmented into small patches and the patches with large image quality differences are chosen for face anti-spoofing to get more discriminative features in [3]. Image quality based methods improve the generalization of face anti-spoofing, but it is still difficult for them to accurately recognize high-definition matte photo and display attacks.

Vital signals based methods One of the obvious differences between live faces and spoof faces is that live bodies have vital signals such as heartbeats, blood flow, eye blinking, facial muscle involuntary micromovements, lips movements and so on. The conditional random field is utilized to distinguish whether there is eye blinking in the input image sequence or not [39]. Such methods learn discriminative features for print attacks, but they will treat display and mask attacks as live faces because display and mask attacks also can do eye blinking as live faces. Live faces have three-dimensional structure but photo faces and display faces have two-dimensional planar structure. The differences of structure lead to the differences in optical flow fields generated by movements of live faces and spoof faces. Kollreider et al. [27] extract the differences of optical flow field data to do face anti-spoofing. It needs a large amount of calculation to compute the optical flow features. Some simulated micromotion of spoof faces may confuse this kind of algorithms. Remote photoplethysmography (rPPG) does non-contact monitoring of pulse only using a webcam [40]. rPPG signals are extracted to detect whether the face in front of the camera has a normal blood volume change resulted in heartbeat or not for live and spoof face classification [21, 33, 37]. Similarly, Wang et al. [49] analyze skin blood flow change from the R and G channels of the input VIS face images to judge whether the face images are captured from a live face or not. However, extracting pulse or skin blood flow change from the images captured by a web camera is greatly affected by the illumination and motion of faces, so the robustness of such face anti-spoofing methods needs to be enhanced.

Deep learning based methods Excepting hand-crafted descriptors, CNN is also utilized to learn discriminative and generalized features from input VIS images automatically. Because it is difficult to make a large number of spoof faces, the data amount of the public face anti-spoofing dataset is usually much lower than that of the datasets of other areas of face recognition. Over-fitting and generalization of features are the key issues to be considered. Yang et al. [36, 53] utilize specially designed convolution neural networks to learn deep features from different scales of face images and feed the learned features to the SVM classifier for live and spoof face classification. They exploit the features learning ability of convolution neural network and meanwhile alleviate the effect of the over-fitting problem via training an SVM classifier. Li et al. [48] fine-tune the classification model which is pre-trained on other big datasets of face recognition to get novel face anti-spoofing classification model. It leverages the discrimination given by the other big datasets. Principle component analysis is utilized to reduce the dimensionality of convolutional features learned by a fine-tuned VGG-face model to further avoid the over-fitting problem [28]. Atoum et al. [5] first translate the images from RGB color space to HSV and YCbCr color spaces, and then extract local and holistic features from the converted images via a two-stream CNN-based architecture to get more discriminative features. Auxiliary supervisions which include the depth maps and rPPG signals are proposed to guild the learning of the

convolution neural network [34]. It combines the differences in structures and vital signals to improve classification performance. Jourabloo et al. [23] decompose a spoof face image to a spoof noise and a live face image, and then perform live and spoof face classification based on the spoof noise. The visualization of the spoof noise helps to have an intuitive understanding of the differences between live and spoof face images. LBP, LBP-TOP, etc. hand-crafted descriptors and CNN are combined to extract effective features in [16, 31]. Excepting 2D convolutional neural network, 3D convolutional neural network is utilized to extract spatial and temporal information for live faces and spoof face classification in [29]. The learned spatial and temporal information is benefit to improve the performance of face anti-spoofing, but the extraction of temporal information makes 3D CNN based face anti-spoofing methods need frame sequences as input. 2D CNN based methods can get classification results based on a single frame, which requires less calculation and computation time. Our approach utilizes 2D CNN to extract fusing features from VIS and NIR image pairs instead of single type VIS or NIR images, combining the clues in both VIS and NIR images for face anti-spoofing.

Other hardware based methods Excepting utilizing VIS images captured by common cameras, researchers also explore the clues in the images captured by other hardware such as near-infrared cameras [25, 45, 54, 57], short wave infrared cameras [44], thermal infrared cameras [24], multi-spectrum cameras [42], light field cameras [41] and deep cameras [50] for face anti-spoofing. We also utilize the clues in near-infrared images, so we review the near-infrared images based methods in detail. Similar to VIS images based methods, Gabor descriptor is utilized to extract texture information from NIR images for live and spoof face classification [54]. The experiments show that it works perfectly to anti-spoof print attacks on their self-collected dataset. The reflection disparity under 850 and 685 nm illuminations is exploited to distinguish live face skins from fake materials in [25]. Zhang et al. [57] analyze reflection properties of human skin versus non-skin from multi-distance NIR images via using the Lambertian model. The use of multi-distance NIR images enables this method to be less sensitive to the distance between the used camera and target being detected. Sun et al. [45] extract the reflection information of face region and context information around face region from the near-infrared differential images. The constructed near-infrared differential images not only contain rich reflection differences but also avoid uncontrolled illumination changes. These near-infrared images based methods all take advantage of the significant reflection differences in NIR images for face anti-spoofing, but they all need near-infrared cameras to capture NIR images, especially some of them need custom-built near-infrared cameras with controllable active near-infrared lights to construct near-infrared differential images. Our approach utilizes an image translation model to generate corresponding NIR images for live and spoof face images without adding any near-infrared cameras to capture NIR images.

2.2 Image translation

Image translation is a constrained image generation process which maps an input image to a desired output image. It is widely applied in the field of face image synthesis, such as face aging [4], face frontalization [47] and face sketch synthesis [15, 56]. Take the studies in face sketch synthesis as example, Zhang et al. [56] jointly use sparse representation based greedy search, multi-feature-based optimization model, and cascaded image regression to generate sketches with specific style for the input photos. A perceptual metric, namely Scoot, is proposed to measure perceptual similarity for face sketch synthesis according to human choice

principles in [15]. The proposed Scoot integrates both spatial structure and co-occurrence texture, which help the Scoot to obtain more similar results to human perception than previously widely used metrics. In recent years generative adversarial network [19] is one of the most popular research topic in computer vision. It has excellent ability in realistic image translation. Pix2pix [22] introduces conditional GAN as a general and promising solution for image translation. It utilizes adversarial loss and L1 loss to constrain the process of learning a mapping from paired source and target images. Considering the inconvenience of collecting paired images, CycleGAN [60], DualGAN [55], DiscoGAN [26] utilize a cycle consistency loss to supervise model learning the mapping from source style to target style without paired images. A common representation across different styles is learned for unsupervised image translation in [32]. StarGAN [12] performs multiple style image translation in a single model. It takes advantage of multi-task learning to improve the generalization ability of image translation among multiple domains. In this paper, we aim to utilize the generated NIR images to do face anti-spoofing classification. It means that the generated NIR images (target style) should preserve the live face and spoof face category information of corresponding VIS face images (source style). However, The popular GANs like CycleGAN just aim to translate the source images to similar target style images. It does not promise to preserve the differences among different image categories. The visual qualitative experiment results are shown in Fig. 5a. We can see that the NIR images of live and spoof faces generated by CycleGAN are similar with each other. Some live face images have the near-infrared characteristics of spoof face images. Such generated NIR images will reduce the performance of face anti-spoofing. The proposed MCT-GAN is inspired by CycleGAN, but the MCT-GAN is designed to translate face images from VIS style to NIR style while preserving the category differences among generated images.

3 Proposed method

Our aim is to explore the clues in both VIS images and NIR images to improve the performance of face anti-spoofing while avoiding adding additional near-infrared hardware. As illustrated in Fig. 3, the proposed method consists of two components: multiple categories NIR image generation and fusing features learning, which are introduced in Sections 3.1 and 3.2, respectively.

3.1 Multiple categories NIR image generation

The reflection attributes of live faces and spoof faces are obviously different under near-infrared spectrum. These differences are characterized by the perceptual style differences like color and texture information in NIR images. They are all crucial clues for face anti-spoofing. In this paper, we utilize the generated NIR images to improve the performance of face anti-spoofing. It is worth noting that the generated NIR images can be truly beneficial for face anti-spoofing only if they preserve the corresponding characteristics of real NIR images of live faces and spoof faces while generating. Therefore, the image translation model should be able to generate face images with different near-infrared characteristics according to the category of faces. In order to generate simulated NIR images, we propose a multiple categories image translation generative adversarial network for the translation of live and spoof face images from VIS style to NIR style, as shown in Fig. 3.

We are inspired by the cycle consistency idea of CycleGAN to do multiple categories image translation. We also use two generators G_a , G_b and two discriminators D_a , D_b to

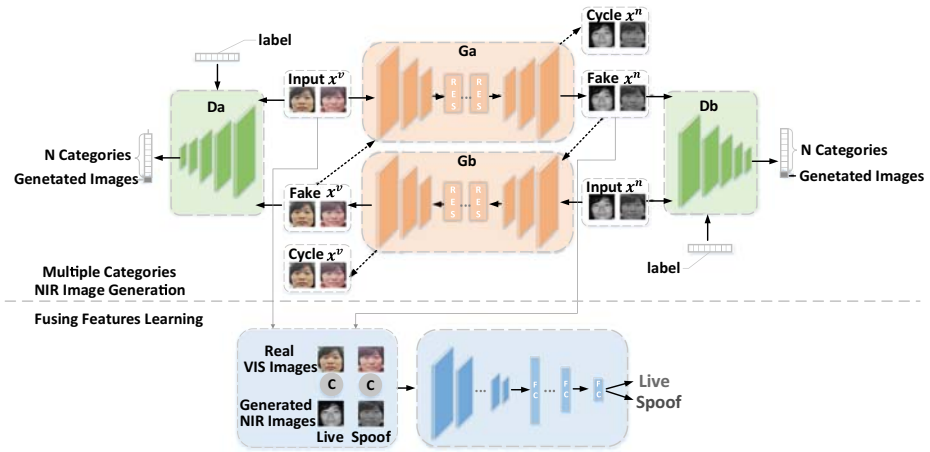


Fig. 3 The overview of the proposed method. It consists of two components: multiple categories NIR image generation and fusing features learning. The proposed MCT-GAN generates the corresponding NIR style images of VIS live and spoof face images firstly, which also preserves the differences between real NIR live and spoof face images during generating. Then, based on paired VIS and generated NIR images, we utilize a CNN to learn fusing features for live and spoof face classification

construct a cycle image translation structure: $x^v \xrightarrow{G_a} x^n, x^n \xrightarrow{G_b} x^v$. G_a translates source VIS style image x^v to target NIR style image x^n . D_a classifies real VIS style image x^v and the generated VIS style image $G_b(x^n)$ into $N+1$ categories. y_r and y_g denote the $N+1$ dimensional label vectors for real input images and generated images, respectively. N is the category number of real input image. In face anti-spoofing task, real input images have two categories: live face and spoof face. Another 1 represents that the image is a real image or a generated image. G_b, D_b resemble G_a, D_a , except the input image is real NIR style image x^n and generated NIR style image $G_a(x^v)$. G_a and G_b seek to distinguish the category of the input image and generate the image of corresponding style. G_a and G_b try to make discriminators be unable to distinguish the generated images from real input images but classify them to corresponding category of their real input images. D_a and D_b seek to distinguish the generated images from real input images and classify the real input images into the N categories. The adversarial loss functions are defined as:

$$L_{ganG_a} = E_{x^v \sim P_{data}(x^v), y_r \sim P_y(y_r)} \|y_r - D_b(G_a(x^v))\|_2, \quad (1)$$

$$L_{ganG_b} = E_{x^n \sim P_{data}(x^n), y_r \sim P_y(y_r)} \|y_r - D_a(G_b(x^n))\|_2, \quad (2)$$

$$L_{ganD_b} = E_{x^n \sim P_{data}(x^n), y_r \sim P_y(y_r)} \|y_r - D_b(x^n)\|_2 \\ + E_{x^v \sim P_{data}(x^v), y_g \sim P_y(y_g)} \|y_g - D_b(G_a(x^v))\|_2, \quad (3)$$

$$L_{ganD_a} = E_{x^v \sim P_{data}(x^v), y_r \sim P_y(y_r)} \|y_r - D_a(x^v)\|_2 \\ + E_{x^n \sim P_{data}(x^n), y_g \sim P_y(y_g)} \|y_g - D_a(G_b(x^n))\|_2. \quad (4)$$

We also adopt the cycle loss for ensuring the consistency of the real VIS image x^v and the cyclically generated VIS image $G_b(G_a(x^v))$ processed by the cycle generation structure

which consists of G_a and G_b , similarly to real NIR image x^n and the cyclically generated NIR image $G_a(G_b(x^n))$. The loss function is defined as:

$$L_{cyc} = E_{x^v \sim P_{data}(x^v)} \|x^v - G_b(G_a(x^v))\|_1 + E_{x^n \sim P_{data}(x^n)} \|x^n - G_a(G_b(x^n))\|_1. \quad (5)$$

We aim to ensure that the generated NIR images are as close as possible to the real NIR images. In this case, generated NIR images will work effectively as real NIR images for face anti-spoofing. In theory, if the pixel values of two images are the same, then these two images can be regarded as the same. Therefore, we use pixel-wise similarity to restrict the consistency of the generated image and real target style image, which are given by:

$$L_{pva} = E_{x^v \sim P_{data}(x^v), x^n \sim P_{data}(x^n)} \|x^n - G_a(x^v)\|_1, \\ L_{pvb} = E_{x^v \sim P_{data}(x^v), x^n \sim P_{data}(x^n)} \|x^v - G_b(x^n)\|_1. \quad (6)$$

Real NIR images provide obvious differences for face anti-spoofing classifier to maximize the distance between features of live faces and spoof faces. Our face anti-spoofing classifier utilizes convolution neural network to learn features from generated NIR live and spoof face images. The distance between features of generated NIR live and spoof faces images is expected as far as real NIR live and spoof faces images. If features of real and generated images are the same, the classification results predicted by the classifier will be the same. Therefore, consistency of the features learned from the same style real and generated images is benefit for our face anti-spoofing classification. We exploit a discriminable feature loss to enforce the representation on the penultimate layer of discriminators of the real target image and the generated image to be consistent. $f(x)$ is defined as the representation of image x . The discriminators of MCT-GAN classify the input real images into live face and spoof face two categories. Features level constraint of real and generated images will encourage the generators to generate images preserving the category information of live faces and spoof faces. The loss functions are shown as:

$$L_{dfa} = E_{x^v \sim P_{data}(x^v), x^n \sim P_{data}(x^n)} \|f(x^n) - f(G_a(x^v))\|_1, \\ L_{dfb} = E_{x^v \sim P_{data}(x^v), x^n \sim P_{data}(x^n)} \|f(x^v) - f(G_b(x^n))\|_1. \quad (7)$$

Different categories of NIR face images have different perceptual characteristics (e.g. color, texture, reflection). These characteristics are important clues for face anti-spoofing, so we encourage the generators to generate corresponding NIR style images for live and spoof faces from a perceptual characteristic standpoint. We employ a style loss [18] to encourage the style of generated target images to match the style of real target images. We still use the penultimate layer of discriminator to get the representation. We denote $f(x)$ as the representation of image x . Its shape is $C \times H \times W$. Gram matrix is denoted as $Gr(x) = \frac{1}{C \times H \times W} f(x)f(x)$ to compute the feature correlations for the representation $f(x)$. The style loss functions are:

$$L_{stylea} = E_{x^v \sim P_{data}(x^v), x^n \sim P_{data}(x^n)} \|Gr(x^n) - Gr(G_a(x^v))\|_1, \\ L_{styleb} = E_{x^v \sim P_{data}(x^v), x^n \sim P_{data}(x^n)} \|Gr(x^v) - Gr(G_b(x^n))\|_1. \quad (8)$$

The full objective functions to optimize the G_a , G_b , D_a , D_b are:

$$G_a = L_{ganG_a} + L_{cyc} + L_{pva} + L_{dfa} + L_{stylea}, \\ G_b = L_{ganG_b} + L_{cyc} + L_{pvb} + L_{dfb} + L_{styleb}, \\ D_a = L_{ganD_a}, \\ D_b = L_{ganD_b}. \quad (9)$$

Our source VIS style and target NIR style images both contain different categories of images. For facilitating the generators distinguish the category of the input image and learn the mapping of different categories of images from style to style, we utilize paired VIS and NIR images of live and spoof faces as input data to train MCT-GAN. We adopt the network architectures in [60] for generators and discriminators. The network of generators consists of three convolution layers, nine residual blocks, two fractional stride convolutional layers and one convolutional layer. Every convolutional layer is followed by instance normalization and ReLU layer. Every residual block contains two convolutional layers. The network of discriminators consists of five convolutional layers. The last convolutional layer produces $N+1$ dimensional output. Every convolutional layer is followed by instance normalization except the first and the last layers. LeakyReLU layer is used to every layer except the last layer. G_a , D_b , G_b , D_a are optimized iteratively in sequence when we train our NIR image generation model.

3.2 Fusing features learning

VIS images have rich color and texture information. These clues have been proved that they can improve the performance of face anti-spoofing. The color and texture information in NIR images is not as rich as VIS images, but NIR images preserve obvious reflection differences, which are also import clues for the performance improvement of face anti-spoofing. In order to take full advantage of clues in VIS and NIR images for face anti-spoofing, we propose to learn fusing features based on paired VIS and NIR images, combining the strengths of these two types of images.

We aim to learn a mapping Φ , which maps paired VIS and generated NIR images $\langle x^v, x^{gn} \rangle$ to their live or spoof face label y . We utilize a convolution neural network to learn fusing features from the image pairs. The network structure is shown in Fig. 4. In order to learn fusing features from two types of images, we concatenate the VIS and generated NIR images by image channel as input data for the following network. We fine-tune an ImageNet pre-trained ResNet50 to learn features from the input image pairs for distinguishing spoof face images from live face images.

It is well known that sample acquisition conditions (e.g. illumination conditions, camera type, spoofing medium, printer type, image resolutions, digital device type) have a great

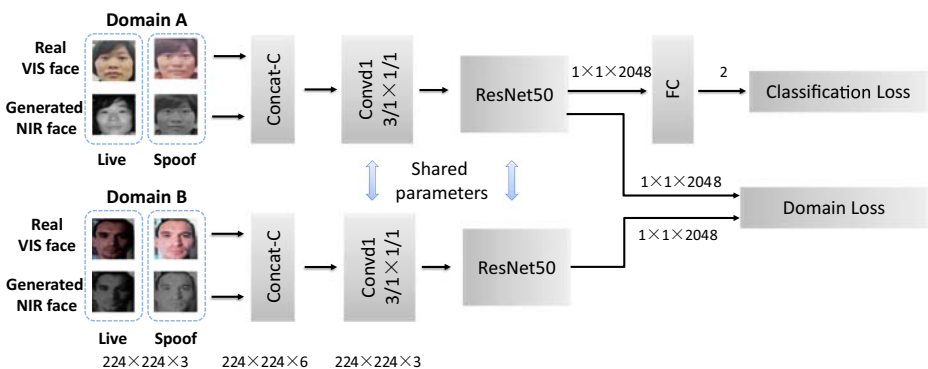


Fig. 4 The network of fusing features learning

influence on the performance of face anti-spoofing. The performance degrades significantly when the training and testing samples are collected under different acquisition conditions (cross-database scenario). This phenomenon is mainly caused by the data distribution discrepancy between training and testing data, which is also called domain shift. The model learned in training set can not generalize well in testing set. Unsupervised domain adaptation aims to learn a discriminative model to reduce the domain shift between two domains A and B while samples of domain A $\{x_{ai}\}_{i=1}^{n_a}$ all have labels $\{y_{ai}\}_{i=1}^{n_a}$ but the samples of domain B $\{x_{bi}\}_{i=1}^{n_b}$ are unlabeled. n_a, n_b are the number of the samples of domain A and domain B , respectively. The face anti-spoofing task under the cross-database scenario can be regarded as an unsupervised domain adaptation task. In order to further improve the generalization of learned features, we exploit classification loss L_e and domain loss L_d to collaboratively supervise the fusing features learning.

$$L_e = - \sum_{c=1}^N y_{ai}^c \log \Phi(< x_{ai}^v, x_{ai}^{gn} >)^c \quad (10)$$

$$L_d = \|\mu_k(h(< x_a^v, x_a^{gn} >)) - \mu_k(h(< x_b^v, x_b^{gn} >))\|_{F_k}^2 \quad (11)$$

$$K := \left\{ k : k = \sum_{u=1}^m \beta_u k_u, \sum_{u=1}^m \beta_u = 1, \beta \geq 0, \forall u \in \{1, \dots, m\} \right\} \quad (12)$$

As mentioned above, the input of our fusing feature learning network is VIS and generated NIR image pair $< x^v, x^{gn} >$. We adopt cross entropy loss as the loss function of live and spoof face classification for input image pair $< x_{ai}^v, x_{ai}^{gn} >$ of domain A , as shown in (10). The domain loss L_d , defined as (11), is utilized to match the feature distribution of domain A and domain B . We introduce the multiple kernel maximum mean discrepancies (MK-MMD) [20] as a discrepancy metric between two domains. F_k denotes a reproducing kernel Hilbert space (RKHS) with reproducing kernel k . $h(< x_*^v, x_*^{gn} >)$ is the feature distribution of domain A or domain B . We choose the output of the last pool layer of ResNet50 as the features to be transferred. The mean embedding of $h(< x_*^v, x_*^{gn} >)$ in F_k is a unique element $\mu_k(h(< x_*^v, x_*^{gn} >)) \in F_k$. The multi-kernel k is a weighted summation of m different kernels $\{k_u\}$ with corresponding coefficients $\{\beta_u\}$. The domain loss L_d is the RKHS-distance between the mean embeddings of $h(< x_a^v, x_a^{gn} >)$ and $h(< x_b^v, x_b^{gn} >)$.

The parameters of the fusing features learning model are learned by minimizing L_{sf} , which is defined as

$$L_{sf} = L_e + \lambda L_d \quad (13)$$

Minimizing the classification loss ensures that the model learns discriminative features for the live and spoof face classification. Minimizing the domain loss makes the model learn domain-invariant features for the input two domains. The model tries to map images of two domains to similar feature distribution so that these two domains are indistinguishable during training. λ is a coefficient to balance the two objectives for desired discriminative

and generalized fusing features learning. It is updated dynamically with the training epoch increasing according to (14). λ_i denotes the coefficient λ of the i th epoch. N_{epochs} is the training times of the fusing features learning.

$$\lambda_i = 2/(1 + e^{(-10*i/N_{epochs})}) - 1 \quad (14)$$

Algorithm 1 The training procedure of the face anti-spoofing method with generated NIR images.

Input: n_p labeled VIS and NIR image pairs $\{(\langle x_i^v, x_i^n \rangle, y_i)\}_{i=1}^{n_p}$, n_a labeled VIS images $\{(x_{ai}^v, y_{ai})\}_{i=1}^{n_a}$ of domain A, n_b unlabeled VIS images $\{x_{bi}^v\}_{i=1}^{n_b}$ of domain B.

Output: Well-trained NIR image generation model G_a^* , fusing features learning model Φ^* .

- 1: Update the parameters of G_a , D_b , G_b , D_a to minimizing corresponding objective function of G_a , D_b , G_b , D_a in (9) iteratively in sequence by the adaptive moment estimation.
 - 2: Generate corresponding NIR image for each single VIS image in $\{x_{ai}^v\}_{i=1}^{n_a}$ and $\{x_{bi}^v\}_{i=1}^{n_b}$: $x_{ai}^{gn} = G_a(x_{ai}^v)$; $x_{bi}^{gn} = G_a(x_{bi}^v)$.
 - 3: Learn the mapping Φ from the VIS and generated NIR image pairs $\{(\langle x_{ai}^v, x_{ai}^{gn} \rangle, y_{ai})\}_{i=1}^{n_a}$, $\{(\langle x_{bi}^v, x_{bi}^{gn} \rangle, y_{bi})\}_{i=1}^{n_b}$ by applying the stochastic gradient descent to update the parameters of Φ and minimize the loss function L_{sf} .
 - 4: Output $G_a^* = G_a$, $\Phi^* = \Phi$.
-

4 Algorithm flow

We have introduced our face anti-spoofing method which leverages the discriminative clues in both visible light images and near-infrared images, and meanwhile avoids utilizing any near-infrared equipment during testing. The training and application procedures of the proposed method are summarized in Algorithm 1 and Algorithm 2, respectively. The proposed method first trains a NIR image generation model based on the proposed MCT-GAN to take the place of near-infrared equipment. Then, it translates the VIS images, which will be used to train the fusing features learning model, to NIR images with corresponding category style by the trained NIR image generation model. Finally, the proposed method trains a fusing features learning model based on the VIS and generated NIR images to do live and spoof face classification. In the application procedure, the proposed method first translates the captured VIS image to a NIR image via the trained NIR image generation model, and then feeds the captured VIS image and generated NIR image to the trained fusing features learning model to predict class label for the captured VIS image.

Algorithm 2 The application procedure of the face anti-spoofing method with generated NIR images.

Input: Unlabeled single VIS image x^v , well-trained NIR image generation model G_a , fusing features learning model Φ .

Output: prediction of class label y' : live or spoof face.

- 1: Generate NIR image x^{gn} for the captured VIS image: $x^{gn} = G_a(x^v)$.
 - 2: Output $y' = \Phi(\langle x^v, x^{gn} \rangle)$.
-

5 Experiments

5.1 Experimental setup

Database We do experiments to evaluate our method qualitatively and quantitatively. For qualitative evaluation, we show visualized results on CASIA-MFSD, Replay-Attack and a dataset collected by ourselves. For quantitative evaluation, we evaluate the face anti-spoofing performance on CASIA-MFSD and Replay-Attack datasets. For fair comparison with previous methods, our experiments use the official overall test protocols of these two datasets.

Experimental setting We collect VIS and NIR frame pairs to train the NIR image generator model and test the model qualitatively. We choose 31 participants to record data, twenty-six of them for training set, five other participants for qualitative testing set. We take three different digit photos for every participant by a Nikon D90 Single-lens reflex camera. All digit photos are printed into chromatic photos on A4 paper by a Cannon imageCLASS MF249dw printer. A visible light and near-infrared binocular camera is used to record the VIS and NIR frame pairs for the live accesses and print attacks. We record the frames in two office lighting conditions. For every light condition, we record 1000 VIS and NIR frame pairs for every recorded object. The recorded object is required to move back and forth at a distance of 30cm to 60cm from the camera. The wavelength of the near-infrared camera is 850nm. All frames are saved as 600*480 resolutions. Figure 2 shows some examples of the collected image pairs.

Face detector in [58] is utilized to detect face for all frames in the collected dataset, CASIA-MFSD and Replay-Attack. The detected face images are all resized to 256*256. We use Tensorflow [1] to train and test our image generation model. The parameters of MCT-GAN are listed follow: the category number is 2, the learning rate is 0.0002, the resolutions of the generated NIR image is 256*256. We use PyTorch to train and test the fusing features learning model. The fusing features learning model is trained via using SGD with a mini-batch size of 32, weight decay of 0.0005, momentum of 0.99. The learning rate starts from 0.01 and decreases exponentially with the training epoch increasing.

Evaluation metrics To compare with previous methods, we use the EER and HTER [11] as the evaluation. FRR is the false reject rate. FAR is the false accept rate. EER is computed on development set. It is half of the sum of FRR and FAR when FRR equal to FAR. t is the corresponding threshold when FRR equal to FAR. HTER is defined as (15). CASIA-MFSD does not have development set, so we only give the EER which is computed on testing set.

$$HTER = \frac{(FRR(t, D) + FAR(t, D))}{2} \quad (15)$$

5.2 Component analysis

The performance of face anti-spoofing depends on whether the generated NIR images preserve enough categories information or not. Therefore, we first investigate the role of the proposed architectures and loss functions of MCT-GAN for the proposed method. We compare 8 models with different architectures and loss functions combinations, i.e., w/o pixel value loss L_{pv} , discriminative feature loss L_{df} and style loss L_{style} , w/o discriminative feature loss L_{df} and style loss L_{style} , w/o pixel value loss L_{pv} and style loss L_{style} , w/o

pixel value loss L_{pv} and discriminative feature loss L_{df} , *w/o* style loss L_{style} , *w/o* discriminative feature loss L_{df} , *w/o* pixel value loss L_{pv} , and proposed method(MCT-GAN+FFL) in each case. FFL denotes the fusing features learning model.

For each model, we utilize VIS and NIR face image pairs from the training set of collected dataset to train corresponding NIR image generation model. Then we feed all detected face images of CASIA-MFSD and Replay-Attack datasets including live face images, print attack face images and display attack face images in training set, development set and testing set to NIR image generation model for generating corresponding NIR images. After that, we feed the source VIS images and the corresponding generated NIR images in training set, development set and testing set to the proposed fusing feature learning model to get EER and HTER for each dataset.

Table 1 shows the face anti-spoofing results utilizing the NIR images generated by different models. We can see that the proposed method with MCT-GAN achieves the best performance for face anti-spoofing. The proposed architectures, pixel value loss L_{pv} , discriminative feature loss L_{df} and style loss L_{style} collaboratively give different categories of VIS images different NIR characteristics, which are beneficial for improving the face anti-spoofing performance. From the experiment results, we can observe that without one of them, the performance will decline. For example, the performance of the model trained without L_{pv} is decreased by 25.00% and 48.26% on Replay-Attack dataset and CASIA-MFSD dataset, respectively. The performance of the model trained without L_{style} is decreased by 53.13% and 26.75% on Replay-Attack dataset and CASIA-MFSD dataset, respectively. The L_{pv} , L_{df} and L_{style} constrain the generated NIR images to be enough close to the real NIR images from three different levels (local pixel values, global semantic features, and image style). The roles played by the L_{pv} , L_{df} and L_{style} can be regarded as complementary for generating realistic and discriminative NIR images to improve the performance of face anti-spoofing.

The pixel value loss L_{pv} constrains the generated NIR images to be enough close to the real NIR images on the pixel values. Pixel-by-pixel comparison is helpful to promote the similarity on local pixels of two images. However, the pixel value loss L_{pv} does not consider the image quality (e.g., perceptual quality, textures), which usually results in the loss of high frequency details and unsatisfied perceptual quality in the generated NIR images. The discriminative feature loss L_{df} controls generators to generate different categories of images at discriminative features level. It measures the global semantic differences between

Table 1 Comparison of different components on CASIA-MFSD and Replay-Attack datasets

Model	Replay-Attack		CASIA-MFSD
	EER (%)	HTER (%)	EER (%)
<i>w/o</i> L_{pv} , L_{df} and L_{style}	1.89	1.05	2.91
<i>w/o</i> L_{pv} and L_{style}	1.88	1.33	3.61
<i>w/o</i> L_{df} and L_{style}	2.65	1.07	3.40
<i>w/o</i> L_{pv} and L_{df}	3.09	1.02	3.27
<i>w/o</i> L_{pv}	2.37	0.60	3.44
<i>w/o</i> L_{df}	2.08	0.85	3.43
<i>w/o</i> L_{style}	2.05	0.96	2.43
Proposed method (MCT-GAN+FFL)	1.84	0.45	1.78

the real and generated NIR images. Compared to the pixel value loss L_{pv} , the discriminative feature loss L_{df} demands that the generated NIR image is perceptually similar to the real NIR image rather than strictly forcing every pixel value at the corresponding positions of the real and generated NIR images to match. The discriminative feature loss L_{df} is favorable to generating visually more perceptible NIR images and further improving the performance of face anti-spoofing. From the experiment results in Table 1, we can see that the discriminative feature loss L_{df} contributes a lot to improve the face anti-spoofing performance on both two datasets. The style loss L_{style} constrains the similarity between the real NIR image and the generated NIR image on image styles like colors and textures. It helps to produce NIR images with more realistic textures and higher perceptual quality, which also are discriminative clues for live and spoof face classification.

The focus of the pixel value loss L_{pv} , discriminative feature loss L_{df} and style loss L_{style} is different and single, but all of their focuses are important factors for effective NIR image generation. Unsatisfactory results are obtained when using some of them alone. The proposed method integrates the advantages of three complementary loss functions to boost the performance of face anti-spoofing.

It is worth to note that the importance of the three loss functions for performance improvement is different on the CASIA-MFSD dataset and Replay-Attack dataset. Among the three kinds of loss functions, the removal of L_{pv} has the least effect on the performance on Replay-Attack dataset. Unlike Replay-Attack dataset, the removal of L_{style} has the least effect on the performance on CASIA-MFSD dataset. Jointly using L_{pv} , L_{df} and L_{style} achieves good performance on both datasets. As mentioned earlier, the training set and testing set of the NIR image generation models are from different datasets. Generating NIR images for CASIA-MFSD and Replay-Attack datasets tests the generalization performance of the NIR image generation models. Therefore, the results also indicate that jointly using the three loss functions to guide the model optimization can improve the generalization performance of the proposed NIR image generation model.

5.3 Quantitative comparison of CycleGAN and MCT-GAN

We compare the generated NIR images by CycleGAN and MCT-GAN in terms of face anti-spoofing performance improvement. Similar to the experimental procedures introduced before, We first use the same training set to train CycleGAN and MCT-GAN models. Then we translate all VIS face images in CASIA-MFSD and Replay-Attack datasets to corresponding NIR face images by trained CycleGAN and MCT-GAN models, and utilize the source VIS face images and generated NIR face images to train and test the fusing features learning model. From the comparison results presented in Table 2, it can be noticed that the NIR images generated by MCT-GAN make the live and spoof face classifier obtain a HTER of 38.36% improvement on Replay-Attack dataset and an EER of 39.46% improvement on CASIA-MFSD dataset compared to CycleGAN, respectively. Performance improvement of

Table 2 Comparison of CycleGAN and MCT-GAN on CASIA-MFSD and Replay-Attack datasets

Model	Replay-Attack		CASIA-MFSD
	EER (%)	HTER (%)	EER (%)
CycleGAN + FFL	3.45	0.73	2.94
Proposed method (MCT-GAN+FFL)	1.84	0.45	1.78

face anti-spoofing indicates that MCT-GAN generates more discriminative NIR face images for live and spoof face classification than CycleGAN.

5.4 Visualization and analysis

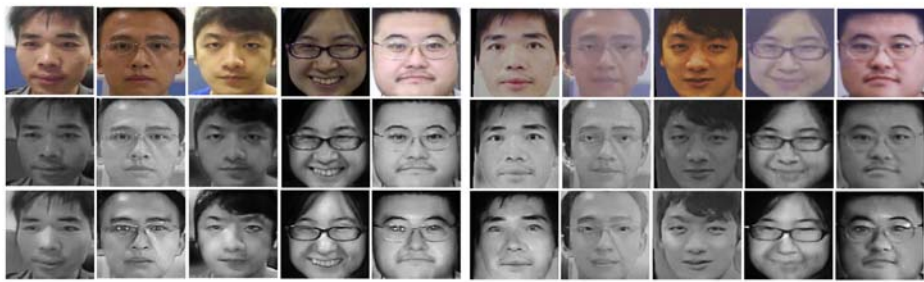
We evaluate the generated NIR images of MCT-GAN and CycleGAN via visualization analysis. We feed the collected qualitative testing set to MCT-GAN and CycleGAN models trained above to get the visualize comparison among the NIR images captured by camera, NIR images generated by MCT-GAN and CycleGAN. Fig. 5a shows some example results. There are some offsets between the VIS and NIR face images recorded by the visible light and near-infrared binocular camera. Neither CycleGAN nor MCT-GAN do any spatial transform. Therefore, generated NIR images generally have the same face structure information as the source input images. In other word, generated NIR images also have some offsets with real NIR images. We can observe that CycleGAN, to a great extent, does not distinguish the category of the source VIS style images. CycleGAN gives near infrared characteristics of print attack face images to real access face images and print attack face images. The loss of near infrared differences is not conducive to subsequent face anti-spoofing task. MCT-GAN succeeds in doing multiple categories image translation from source VIS style to target NIR style. It gives live and spoof face images corresponding near infrared image characteristics, e.g., the color of eyes, the color and texture of face skin. The category differences generated by MCT-GAN are important classification clues for face anti-spoofing.

We also utilize CASIA-MFSD and Replay-Attack datasets as testing set of MCT-GAN and CycleGAN to do visualize comparison of generated NIR images. Figure 5b and c show some example results. From Fig. 5a, b, and c, we can see that MCT-GAN generates more realistic NIR images than CycleGAN. Lots of images generated by CycleGAN have bright cavity. It is worth mentioning that the training set and testing set of MCT-GAN and CycleGAN are from different datasets in these experiments. The training set is from the collected dataset. The testing sets are CASIA-MFSD dataset and Replay-Attack dataset. It is well known that training set and testing set from different datasets tests the generalization ability of the model. MCT-GAN obtains better NIR image generation results on CASIA-MFSD and Replay-Attack datasets, which also demonstrates that MCT-GAN has stronger cross datasets generalization ability than CycleGAN.

We also conduct visualization experiments to compare NIR images generated by MCT-GAN with NIR images captured by NIR equipment. The qualitative testing set contains NIR images captured by NIR equipment. Therefore, we utilize the qualitative testing set to do the visualization analysis. The second and third rows of the Fig. 5a show examples of captured NIR images and generated NIR images, respectively. As mentioned earlier, the generated live and spoof NIR images have near infrared image characteristics of corresponding category to some extent. For example, the skin of live faces looks more smooth than spoof faces. The eyes of live faces look a little turbid, but the eyes of spoof faces look clear. However, there are also many differences between generated NIR images and captured NIR images. There are some spatial shifts between generated NIR images and captured NIR images as previously described. Spatial shifts existing in the VIS and captured NIR image pairs can add the data variations in camera angle, which are favorable to performance improvement of a live and spoof face classification model. There are no obvious spatial shifts in the VIS and generated NIR image pairs. This is a loss for the goal of performance improvement. In addition, the colors and textures of some generated NIR images are not as smooth and natural as the captured NIR images.



(a)



(b)



(c)

Fig. 5 **a** comparison of the generated NIR images on qualitative testing set. **b** comparison of the generated NIR images on CASIA-MFSD dataset. **c** comparison of the generated NIR images on Replay-Attack dataset. Five columns on left are live face images. Five columns on right are spoof face images. For figure (a), each column from top to bottom are VIS images, NIR images captured by camera, NIR images generated by MCT-GAN and CycleGAN, respectively. For figure (b) and (c), each column from top to bottom are VIS images, NIR images generated by MCT-GAN and CycleGAN, respectively

5.5 Advance of the generated NIR images

We perform testing to evaluate the advantage of the generated NIR images for face anti-spoofing. We compare the performance of only using VIS images as input data as well as

Table 3 Comparison results of only using VIS images as input data as well as using both VIS and generated NIR images as input data on CASIA-MFSD and Replay-Attack datasets

Input data	Replay-Attack		CASIA-MFSD
	EER (%)	HTER (%)	EER (%)
Only VIS images	1.91	1.10	3.25
VIS and NIR images	1.84	0.45	1.78

using both VIS and generated NIR images as input data on two datasets. The model of MCT-GAN trained on the collected dataset is utilized to translate the VIS images to NIR images for CASIA-MFSD and Replay-Attack datasets. The fusing features learning model introduced above is utilized to do live and spoof face classification for the evaluation of using both VIS and generated NIR images as input data. The NIR image data layer and the concatenation layer are removed when we train and test the model of only using VIS images as input data. Table 3 shows the performance comparison of these two type of input data. Generated NIR images help the live and spoof face classifier to achieve a HTER of 59.09% and an EER of 45.23% improvement for Replay-Attack and CASIA-MFSD dataset, respectively. The generated NIR images effectively improve the performance of face anti-spoofing. The results demonstrate that MCT-GAN is able to generate discriminative NIR images for face anti-spoofing. In other word, MCT-GAN distinguishes the category of the input source VIS image and gives it corresponding NIR color and texture style. It is also worth mentioning that the training set and testing set of MCT-GAN in these experiments are different datasets. The effectiveness of generated NIR images also demonstrates the cross dataset generalization ability of MCT-GAN.

The CASIA-MFSD dataset contains a diverse range of attack variations. The dataset records three different image qualities, which are low quality, normal quality and high quality. The spoof faces of CASIA-MFSD dataset consist of warped photos, cut photos and displayed videos. The spoof faces are all made from electronic photos or videos with high quality. We analyze the performance impact of the generated NIR images according to attack categories and image qualities on CASIA-MFSD dataset, respectively. The results are presented in Tables 4 and 5. From an overall perspective, we can see that the generated NIR images help to improve the performance significantly. Nevertheless, for different attack categories or image qualities, the proposed method still has different performance.

Compared to only using VIS images as input, the proposed method can distinguish cut photo attacks from real accesses effectively. The performance of displayed attack recognition is worse than that of other two types of attacks. The training images of the NIR image generation model only include NIR images of live accesses and printed attacks. Generating NIR images for the VIS images of displayed attacks tests the generalization of the NIR generation model. It needs to classify the images of displayed attacks as spoof face images and

Table 4 Comparison results of using different types of images as input data on CASIA-MFSD dataset according to attack categories

	Warped	Cut	Display
	FAR (%)	FAR (%)	FAR (%)
Only VIS images	2.81	4.08	3.04
VIS and NIR images	1.80	0.80	2.68

Table 5 Comparison results of using different types of images as input data on CASIA-MFSD dataset according to image qualities

	Low FRR(%)	Normal FRR(%)	High FRR(%)	Low FAR(%)	Normal FAR(%)	High FAR(%)
Only VIS images	2.61	4.88	2.30	3.82	3.48	2.52
VIS and NIR images	0.42	4.80	0.10	0.11	1.72	3.39

give the near-infrared style of spoof face images (printed attacks) to the generated images. Although the generated NIR images help to improve a little performance for detecting displayed attacks, the NIR image generation of displayed attack images is still a challenge for the generalization ability of the NIR image generation model.

The generated NIR images help to achieve an FRR of 83.90% and 95.65% improvement for detecting live face images with low quality and high quality, respectively. For normal quality live images, the FRRs of using two types of images as input data are relatively close. For low and normal quality spoof face images, the FARs of adding generated NIR images are reduced obviously, i.e., $(1.72-3.48)/3.48=50.57\%$ for normal quality spoof face images. However, the generated NIR images increase the FAR of high quality spoof face images by 34.52%. The quality of the images used to train the NIR image generation model is similar to normal quality defined in the CASIA-MFSD dataset. The results indicate that the NIR image generation model can generalize well to low and normal quality images, but it suffers from the generalization performance for high quality spoof face images. The NIR image generation of high quality spoof face images is worthy of further research.

5.6 Comparison with state-of-the-art methods

Intra-dataset Testing We perform experiments to compare our method with the state-of-the-art methods on CASIA-MFSD and Replay-Attack datasets via intra-dataset testing. Table 6 provides comparison results among our proposed method and the state-of-the-art frame based methods. From Table 6, we can see that our proposed method achieves an

Table 6 Intra-dataset testing results on CASIA-MFSD and Replay-Attack datasets

Method	Replay-Attack		CASIA-MFSD EER (%)
	EER (%)	HTER (%)	
LBP [11]	13.90	13.80	18.20
DOG [59]	-	-	17.0
CNN [53]	4.32	2.68	6.25
Color LBP [7]	0.40	2.90	6.20
Color texture [8]	0.40	2.80	2.10
Ft VGG-Face [28]	8.40	4.30	5.20
Com DPCNN [28]	2.90	6.10	4.50
Deep LBP [31]	0.10	0.90	2.30
Patch& Depth [5]	0.79	0.72	2.67
Proposed method	1.84	0.45	1.78

HTER of 37.50% and an EER of 15.24% improvement for Replay-Attack and CASIA-MFSD datasets, respectively. The method in [5] is somewhat similar to the proposed method. It learns features from VIS image patches and generated depth maps for live and spoof face classification. A live human face has a fine three-dimensional structure. It is hard for two-dimensional spoof faces like printed faces and displayed faces to imitate that. Depth maps record the structural differences between live and spoof faces and they help the method to obtain good results[5]. The proposed method generates NIR images to provide more discriminative inter-class differences. Although the differences between two-dimensional spoof faces and live faces in both depth maps and NIR images are obvious, the differences in depth maps are relatively larger than those in NIR images. The performance improvement achieved by the proposed method reveals that MCT-GAN generates efficient NIR images, and the fusing features learning model learns discriminative features from source VIS images and generated NIR images for the final face anti-spoofing goal.

The method in [31] combines the advantages of convolutional neural network and representative handcrafted feature LBP to do live and spoof face classification. It obtains an EER of 0.1% on Replay-Attack dataset. Compared to previous methods, the EER of our proposed method on Replay-Attack dataset is high. We analyze the misclassified samples, and we find that the live access images of client15 recorded under the controlled scenario lead to this result. Some examples are shown in Fig. 6. We can see that the eye areas of these misclassified images are much darker than other common live face images. This characteristic makes these images look like spoof face images which have low image quality. These images are easily misclassified from the perspective of VIS images. Meanwhile, the differences existing in eye areas of NIR images are important classification clues for the proposed method, but dark eye areas are not conducive to generate corresponding NIR images with the style of live faces. From the angle of NIR images, these images are also easily misclassified as spoof face images by the proposed method. How to enable the NIR image generation model to be robust to this kind of images indicates a future research direction.

Cross-dataset Testing We perform some cross-dataset testing experiments between CASIA-MFSD and Replay-Attack datasets to analyze the performance when training and testing data come from different domains. The cross-dataset testing results in terms of HTER shown in Table 7 demonstrate that the proposed method has good generalization ability compared to the state-of-the-art methods. We denote the evaluation of training the model on CASIA-MFSD dataset and testing on Replay-Attack dataset as protocol CR, and denote the evaluation of training the model on Replay-Attack dataset and testing on CASIA-



Fig. 6 Examples of misclassified samples on the development set of Replay-Attack dataset

Table 7 Cross-dataset testing results between CASIA-MFSD and Replay-Attack datasets

Method	Train	Test	Train	Test
	CASIA-MFSD HTER (%)	Replay-Attack	Replay-Attack HTER (%)	CASIA-MFSD
LBP [13]	55.9		57.6	
Color LBP [7]	47.0		39.6	
CNN [53]	48.5		45.5	
FD-ML-LPQ-FS [6]	50.3		42.6	
Color texture [8]	30.3		37.7	
Noise Modeling [23]	28.5		41.1	
Domain Adapt [30]	27.4		36.0	
Auxiliary [34]	27.6		28.4	
proposed method	22.8		23.4	

MFSD dataset as protocol RC. The proposed method reduces the cross-testing HTER by 16.78% and 17.60% compared to the previous state-of-the-art on protocol CR and RC, respectively. It can be observed that the proposed method obtains generalized features for face anti-spoofing.

The variations of CASIA-MFSD dataset are more diverse than that of Replay-Attack dataset in terms of image qualities, attack categories and acquisition equipment. As a result, the performance of some methods [8, 23, 30] on the protocol RC is worse than the performance on the protocol CR. For example, The HTER of the method in [30] on the protocol RC is 31.39% higher than on the protocol CR. However, the HTER of the method in [34] on the protocol CR is close to the HTER on the protocol RC. The reason lies in that the method leverages generalized classification clues (depth maps and rPPG signals) which are less sensitive to the variations between different datasets in theory. As Table 7 shown, the results of the proposed method on the protocol CR and RC are close. It also reveals that the proposed method learns generalized features from the VIS and generated NIR images across variations between the datasets.

6 Conclusions

In this paper, we propose a novel method for face anti-spoofing, which utilizes the clues in NIR images, but does not need to add additional near-infrared equipment during testing. We first employ a multiple categories image translation GAN to get discriminative near-infrared live and spoof face images. Then, based on paired VIS and generated NIR face images, we utilize a CNN model to learn fusing features for live and spoof face classification. Extensive experiments demonstrate that our method archives excellent results in both realism of the generated NIR face images and face anti-spoofing classification performance.

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